UNIT 2: SHAPES AND REGIONS

SESSION 6: SHAPE MODELS AND SHAPE RECOGNITION

SLO 1: DESCRIBE VARIOUS SHAPE MODELS USED FOR REPRESENTING OBJECT SHAPES IN DIGITAL IMAGES

Word Match

Match the shape model with its correct description.

Shape Model	Description
Boundary Model	A. Represents the shape using the outer contour
Region Model	B. Represents the shape using all pixels inside the object
Skeleton Model	C. Represents the shape using a thin central structure
Graph Model	D. Represents shape using nodes and connections

Write the correct matches:

Boundary Model → A

Region Model → B

Skeleton Model → C

Graph Model → D

Application Thinking

Consider the following applications.

Application	Suitable Shape Model
Character recognition (OCR)	Boundary Model
Medical structure analysis	Skeleton Model
Road network detection	Graph Model